that differs from them dramatically when distance scales get small. Specifically, sub-Planck-scale distances are inaccessible.

As this discussion is quite subtle, let's re-emphasize one central point. If we were to spurn the distinction between "easy" and "hard" approaches to measuring length and, say, continue to use the unwound modes as R shrinks through the Planck length, it might seem that we would indeed be able to encounter a sub-Planck-length distance. But the paragraphs above inform us that the word "distance" in the last sentence must be carefully interpreted, since it can have two different meanings, only one of which conforms to our traditional notion. And in this case, when R shrinks to sub-Planck length but we continue to use the unwound strings (even though they have now become heavier than the wound strings), we are employing the "hard" approach to measuring distance, and hence the meaning of "distance" does not conform to our standard usage. However, the discussion is far more than one of semantics or even of convenience or practicality of measurement. Even if we choose to use the nonstandard notion of distance and thereby describe the radius as being shorter than the Planck length, the physics we encounter—as discussed in previous sections—will be identical to that of a universe in which the radius, in the conventional sense of distance, is larger than the Planck length (as attested to, for example, by the exact correspondence between Tables 10.1 and 10.2). And it is physics, not language, that really matters.

Brandenberger, Vafa, and other physicists have made use of these ideas to suggest a rewriting of the laws of cosmology in which both the big bang and the possible big crunch do not involve a zero-size universe, but rather one that is Planck-length in all dimensions. This is certainly a very appealing proposal for avoiding the mathematical, physical, and logical conundrums of a universe that emanates from or collapses to an infinitely dense point. Although it is conceptually difficult to imagine the whole of the universe compressed together into a tiny Planck-sized nugget, it is truly beyond the pale to imagine it crushed to a point of no size at all. String cosmology, as we shall discuss in Chapter 14, is a field very much in its infancy but one that holds great promise, and may very well provide us with this easier-to-swallow alternative to the standard big bang model.

How General Is This Conclusion?